

Democratizing Precision: High-Performance Hand Gesture Recognition via Low-Cost Exoskeleton and Physics-Informed Graph Neural Networks

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Abstract—Accurate hand gesture recognition is often hindered by the tradeoff between the high cost of commercial data gloves and the unreliability of visual methods. To break this barrier, we introduce a low-cost 3-D-printed exoskeleton glove with a total hardware cost under U.S. \$20. This device accurately measures 15 degrees of freedom (DoF) and achieves high-fidelity dynamic tracking, demonstrated by a Pearson correlation coefficient exceeding 0.98 against desired trajectories. We pair this hardware with a novel angle-graph convolutional network with physical constraints (AGCN-PC), which leverages the glove's precise physical structure as a graph prior and incorporates interjoint coupling constraints. On a self-collected dataset of 12 dynamic gestures from eight subjects, it achieves 95.64% accuracy in a rigorous leave-one-out cross-validation for user generalization, significantly outperforming all baselines. Critically, our model exhibits remarkable data efficiency, achieving over 80% accuracy with only 33% of the training data, a level the baselines fail to reach even with the full dataset. This work validates that the synergy of affordable, high-fidelity hardware and physics-informed artificial intelligence (AI) provides a powerful and accessible path toward democratizing precision gesture recognition.

Index Terms—Graph neural networks (GNNs), hand gesture recognition, high-fidelity sensing, low-cost exoskeleton, physics-informed artificial intelligence (AI).

I. INTRODUCTION

THE human hand, with its remarkable dexterity, serves as a primary interface for interacting with both the physical and digital worlds. Its ability to perform a wide spectrum of tasks, ranging from simple grasps to complex manipulations,

and to convey rich nonverbal information makes it a focal point of scientific inquiry [1], [2]. Consequently, the accurate recognition and interpretation of hand gestures are pivotal for advancing numerous fields, including human–computer interaction (HCI), virtual/augmented reality (VR/AR), robotic teleoperation, and medical rehabilitation [3], [4], [5]. In VR/AR, nuanced gesture control significantly enhances user immersion [6], while in rehabilitative medicine, it enables the quantitative assessment of motor function recovery for patients, such as those recovering from a stroke [7]. Moreover, gesture-based interfaces provide an intuitive control modality for individuals with disabilities, greatly improving assistive technologies [8].

Despite its broad applicability, achieving a high-precision, robust, and accessible gesture recognition system remains a significant challenge. Current technologies primarily fall into two categories: wearable sensor-based methods and computer vision-based methods. Wearable sensor-based approaches have long been a cornerstone of high-fidelity gesture capture. Commercial data gloves, such as the CyberGlove [9] and Manus Prime II [10], integrate high-precision flexural or inertial sensors to provide accurate multidegrees of freedom (DoF) finger motion data [11]. These devices are considered the gold standard in many research applications, and prominent benchmark datasets like NinaPro were collected using them [12]. However, the prohibitive cost of these systems, frequently reaching thousands of dollars, severely limits their accessibility, thereby confining them to specialized research settings and impeding the democratization of precision gesture interaction [13]. To address this cost barrier, numerous studies have explored low-cost alternatives. Approaches using gloves stitched with flexible bend sensors [14] or piezoresistive materials have shown promise, but often struggle with measurement accuracy due to the noncollocation problem, where the sensor's bending axis misaligns with the finger's joint axis [15]. Furthermore, these materials can suffer from signal drift, hysteresis, and sensitivity to environmental factors, necessitating complex and user-specific calibration routines [16], [17]. Other works have utilized arrays of inertial measurement units (IMUs) [18], but these are susceptible to magnetic interference and cumulative drift errors, making stable, absolute joint angle estimation over time a persistent challenge [19].

On the other hand, computer vision-based methods offer a noncontact alternative, using RGB or depth cameras [20] to

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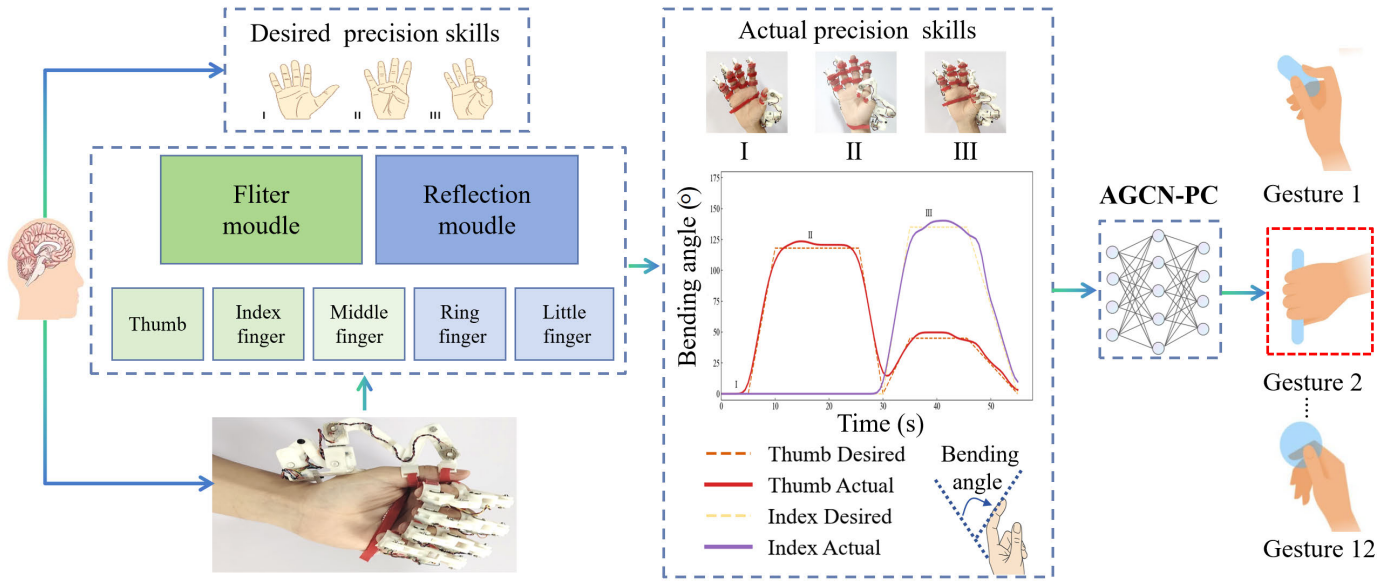


Fig. 1. Overview of the low-cost, high-precision gesture recognition system. Conceptual diagram of the system, illustrating the process from the user's desired precision skills to the actual measurement of finger bending angles, and then to gesture recognition via the AGCN-PC model. It also shows the interaction between fingers and the exoskeleton.

capture hand images. Significant research has focused on deep learning models for hand pose estimation from these images [21], [22]. While these methods benefit from low-cost camera hardware [23], they face inherent limitations in measurement reliability. Performance is often degraded by fluctuating illumination, complex backgrounds, and occlusions—either self-occlusion between fingers or external occlusion by objects [24], [25]. Moreover, state-of-the-art vision algorithms are computationally intensive, posing challenges for real-time deployment on resource-constrained devices [26]. A more fundamental limitation is their reliance on a generic, statistical hand model. This “one-size-fits-all” approach struggles to adapt to the vast individual differences in hand size, shape, and kinematics, thus limiting the precision of joint angle estimation [27].

In recent years, deep learning on graph-structured data has emerged as a powerful paradigm for skeletal action recognition. Models like graph convolutional networks (GCNs) [28] and spatio-temporal GCNs (ST-GCN) [29] have excelled by explicitly modeling the human body's kinematic structure as a graph. However, the efficacy of these models heavily depends on the quality of the input graph topology. When applied to vision-based hand pose estimation, they often inherit the inaccuracies of the estimated skeleton, which is a low-fidelity representation of the true physical structure. Concurrently, the field of physics-informed machine learning (PIML) has gained traction, advocating for the integration of physical laws and domain knowledge into artificial intelligence (AI) models to improve data efficiency and generalization [30]. This principle suggests that directly embedding known physical constraints, such as joint limits or kinematic dependencies, could significantly enhance model performance.

The analysis of existing works reveals a persistent trilemma in gesture recognition: the tradeoff between cost, accuracy, and robustness. Inspired by the potential of hardware–software

co-design to address these challenges, we propose a novel, integrated system for high-performance gesture recognition, the overview of which is illustrated in Fig. 1.

Our system initiates with the user's intent, which is translated into physical finger movements. These movements are precisely captured by our custom-designed low-cost exoskeleton glove. Uniquely, this hardware bridges the gap between high-end commercial systems and low-cost DIY solutions: unlike commercial data gloves [9], [10], [11] that offer high precision but at a prohibitive cost, or flexible sensor-based alternatives that often suffer from signal drift and hysteresis, our rigid linkage design ensures robust, high-fidelity tracking (Pearson correlation coefficient >0.98) with a total hardware cost under U.S. \$20. The glove converts the mechanical motion into a 15-D joint angle data stream. This high-fidelity data, representing the actual precision skills, serves as direct input to our proposed angle-graph convolutional network with physical constraints (AGCN-PC). The AGCN-PC model then processes this spatio-temporal data to perform accurate classification, identifying the performed action from a set of 12 distinct gestures. This closed-loop conceptual diagram highlights our core philosophy: bridging the gap between user intent and system understanding through precise, physically grounded measurement and intelligent, physics-informed modeling.

Our main contributions are as follows.

- 1) *A Low-Cost, High-Fidelity Exoskeleton Glove*: We designed and implemented a novel 3-D-printed exoskeleton glove that provides high-fidelity, 15-DoF kinematic measurements at an extremely low cost of under U.S. \$20. Its three-bar linkage mechanism directly addresses the noncollocation and instability issues common in other low-cost wearable sensors.
- 2) *AGCN-PC*: We propose a novel neural network architecture that uniquely integrates two key innovations. First, it leverages the exoskeleton's precise mechanical

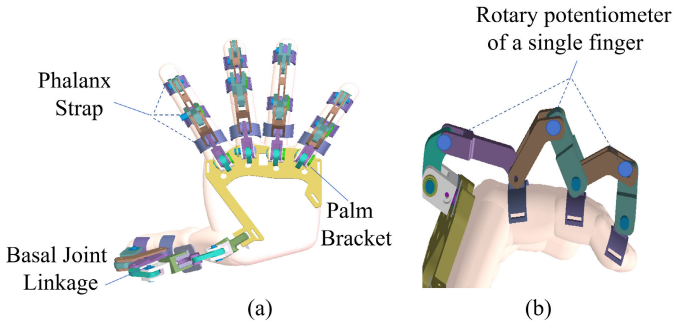


Fig. 2. Mechanical design of the low-cost exoskeleton glove. (a) Overall assembly view, showing the main components. (b) Detailed view of a single finger's three-bar linkage mechanism used for angle sensing. The flexion/extension of the three joints is measured by three separate rotary potentiometers, whose movements are driven by the links attached to the user's finger.

structure to form a *high-fidelity graph prior*, enhanced with physically meaningful *aggregated feature nodes*. Second, it introduces a physics-informed learning objective by explicitly incorporating biomechanical *interjoint coupling constraints*, which act as a powerful regularizer.

- 3) *Comprehensive Validation and Data Efficiency Analysis:* We rigorously validated our system through extensive experiments. Our method achieves 95.64% accuracy in a demanding cross-user validation and a perfect 98.50% accuracy in a user-mixed setting, significantly outperforming all baselines. Critically, we demonstrate the model's remarkable data efficiency, where it achieves over 80% accuracy using only 33% of the training data, a performance level unattainable by baselines even with the full dataset.

II. DESIGN AND METHODS

A. Hardware Design and Sensing Principle

The main structure of the exoskeleton is fabricated using Fused Deposition Modeling 3-D printing with polylactic acid, a common and inexpensive material that allows for rapid prototyping and customization. As shown in Fig. 2(a), the glove consists of three primary components: a rigid dorsal palm mount, individual finger linkage assemblies, and flexible phalanx straps. The dorsal palm mount serves as a stable base, anchoring the entire assembly to the back of the user's hand. Each finger is instrumented with a dedicated linkage assembly responsible for capturing its motion.

The core of our sensing principle lies in the three-bar linkage mechanism designed for each finger, detailed in Fig. 2(b). This mechanism is engineered to translate the complex, multi-joint flexion/extension of a finger into distinct, measurable rotations at the exoskeleton's joints. Specifically, the flexion/extension of the metacarpophalangeal, proximal interphalangeal, and distal interphalangeal joints are mechanically coupled to the rotational axes of the linkage system. This design ensures a direct and repeatable mapping from finger joint angles to sensor rotations, a crucial factor for measurement accuracy.

TABLE I
DETAILED COST BREAKDOWN FOR THE EXOSKELETON GLOVE

Component	Specification	Quantity	Cost (\$)
Angle Sensors	Rotary Potentiometers	15	9.00
3D Printing Material	PLA	50g	4.00
Base Glove	Simple fabric glove	1	3.00
Fasteners & Adhesives	Velcro straps, screws, glue	1 set	2.00
Wiring & Connectors	Dupont wires, pin headers	1 set	2.00
Total Cost			\$20.00

To measure the rotational movements of the linkage mechanism, we integrated low-cost rotary potentiometers at each key revolute joint of the exoskeleton. A total of 15 potentiometers are deployed to capture the primary flexion/extension DoFs of the human hand. For the four fingers (index, middle, ring, and little), three potentiometers each are used to measure the MCP, PIP, and DIP joint angles.

Recognizing the unique and complex kinematics of the thumb, a specialized linkage mechanism, referred to as the Basal Joint Linkage in Fig. 2(a), was designed to capture its crucial motions, including flexion/extension and abduction/adduction at the Carpometacarpal joint, as well as MCP and Interphalangeal joint flexion. This results in a comprehensive 15-DoF sensing capability across the entire hand. User comfort and adaptability are achieved by attaching the 3-D-printed components to the user's hand and finger segments via adjustable Velcro straps. This allows the glove to accommodate a range of hand sizes and ensures that the user can perform gestures naturally without significant hindrance. The interface between the exoskeleton and the human hand is critical for long-term usability. We implemented a fixation system using adjustable Velcro fasteners, which can be micro-adjusted to match the specific thickness of each finger phalanx. This ergonomic consideration ensures that the device remains comfortable and stable even during rapid dynamic gestures. A key objective of this work is to democratize precision gesture sensing. The total hardware cost for a single glove is strictly controlled to be under U.S. \$20, a figure drastically lower than that of commercial data gloves. A detailed componentwise cost breakdown is provided in Table I.

B. Joint Angle Measurement and Calibration

Each rotary potentiometer converts the bending angle of a joint into an analog voltage signal. These voltage signals are collected by NI through an analog-to-digital converter. Since there is typically a nonlinear relationship between the output voltage of the potentiometer and the joint angle, which may also vary due to individual assembly differences, an accurate calibration is the key to obtaining reliable angle data. We adopt a data-driven calibration method. For each sensing unit, we collect a series of voltage outputs under known joint angles. Then, using these paired data points, we fit a mapping model for each sensor. As shown in Fig. 3, it illustrates the fitting relationship between the normalized output voltage and its corresponding joint angle for a single finger sensor. The module converts raw voltage V from sensor i into the calibrated joint angle. We perform 50 cycles of flexion and

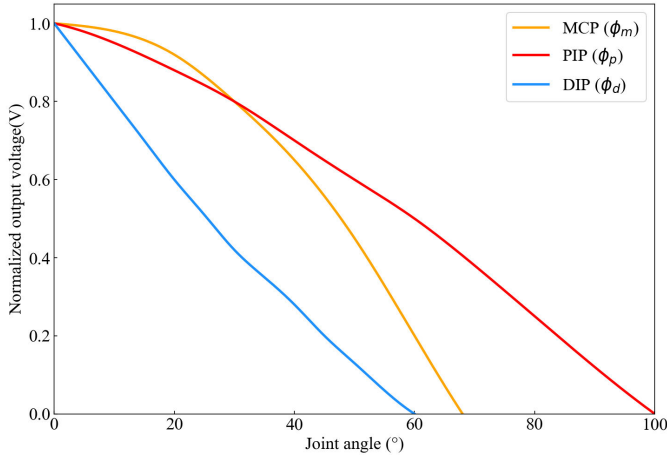


Fig. 3. Normalized output voltage as a function of the corresponding joint angle.

extension for each finger and use the averaged data for model fitting. The collected angle data may be processed with a simple moving average filter to reduce high-frequency noise before being fed into the recognition model.

C. Physics-Constrained GNN Model

1) *Overall Model Architecture*: The proposed AGCN-PC model is built upon the specialized spatio-temporal graph structure illustrated in Fig. 4, and its overall architectural pipeline is depicted in Fig. 5. The model is designed as a sequential pipeline that processes the dynamic angle features through three main stages. First, a spatio-temporal graph module, built upon our novel augmented graph structure, captures the complex spatial dependencies between finger joints at each time step. Second, the spatially enhanced feature sequence is fed into a multilayer bidirectional LSTM network to model the long-range temporal dependencies inherent in dynamic gestures. Finally, the features are passed through a fully connected layer with a specialized loss function that explicitly incorporates our proposed physical constraints, before a final softmax layer outputs the classification probabilities for the 12 gesture classes. We will now detail each of these key components.

2) *Spatio-Temporal Graph With Feature Augmentation*: The foundation of our model is a novel spat-temporal graph representation designed to fully exploit the kinematic information from our exoskeleton. As illustrated in Fig. 4(a), we model the hand at a single time frame as an augmented spatial graph $G = (V, E)$. The node set V consists of two types of nodes: 15 Base Joint Nodes (V_{base}), each corresponding to one of the measured joint angles from the exoskeleton, and five additional aggregated feature nodes (V_{agg}), one for each finger. The feature of an aggregated node is defined as the sum of the angles of all joints within that finger, serving as a powerful, physically meaningful feature representing the overall flexion of each finger. The edge set E is also composed of distinct types, including intrafinger edges that form the core physical graph, virtual interfinger edges to model synergistic movements, and feature aggregation paths connecting the base nodes to their corresponding aggregated node.

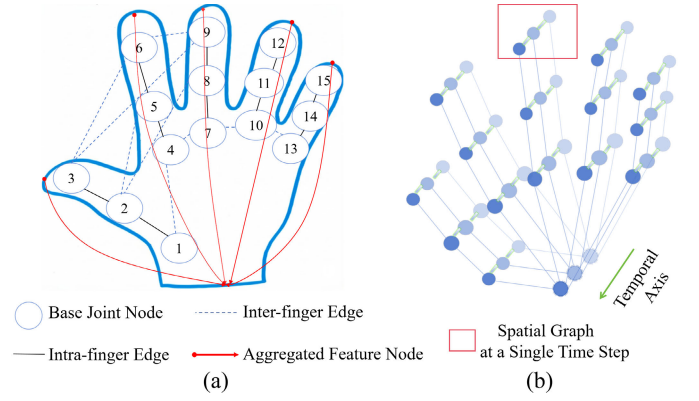


Fig. 4. Architecture of the spatio-temporal graph with feature augmentation. (a) Augmented spatial graph for a single time frame. The graph consists of 15 base joint nodes representing measured angles and five aggregated feature nodes representing the total flexion of each finger. The base nodes within each finger are connected by intrafinger edges, forming the core physical graph. Interfinger edges are added as virtual connections to model synergistic movements between adjacent fingers. The feature aggregation path shows how the aggregated nodes are derived from the base nodes. (b) Full spatio-temporal graph structure. The augmented spatial graph is unrolled along the temporal axis, with corresponding nodes connected across consecutive time steps to model the gesture dynamics.

This augmented spatial graph is then unrolled along the temporal axis, as shown in Fig. 4(b). To learn the spatial dependencies at each time frame, we employ a GCN. For the input features $\mathbf{X}(t) \in \mathbb{R}^{20 \times 1}$ at frame t , the graph convolution at layer l is

$$\mathbf{H}^{(l+1)}(t) = \sigma \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} (\mathbf{A} + \mathbf{I}) \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)}(t) \mathbf{W}^{(l)} \right) \quad (1)$$

where $\mathbf{H}^{(l)}(t)$ is the feature map at layer l , $\mathbf{W}^{(l)}$ is the trainable weight matrix, and $\tilde{\mathbf{D}}$ is the degree matrix of $(\mathbf{A} + \mathbf{I})$. After the GCN module, the feature sequence is processed by a three-layer Bi-LSTM network to capture temporal context, as illustrated in the expanded view in Fig. 5. It is crucial to distinguish the theoretical foundation of our AGCN-PC from generic ST-GCNs [31]. Standard ST-GCNs often operate on skeleton data estimated from visual sensors, which inherently suffer from occlusion-induced noise and nonphysical artifacts. In contrast, AGCN-PC constructs a high-fidelity graph prior directly derived from the exoskeleton's rigid mechanical linkage. By explicitly embedding the interjoint coupling constraints and using a fixed, physically accurate topology, AGCN-PC imposes a strong physical inductive bias. This ensures that the learned features strictly adhere to biomechanical laws, enabling the model to achieve superior generalization and data efficiency compared to purely data-driven approaches that must implicitly learn these physical rules from massive datasets.

3) *Explicit Incorporation of Physical Constraints*: The movements of finger joints are not independent. A well-established biomechanical principle is the strong kinematic interdependence between the proximal (PIP) and distal (DIP) interphalangeal joints during flexion and extension. While early studies often approximated this as a linear relationship, detailed anatomical analyses [32] reveal that this coupling

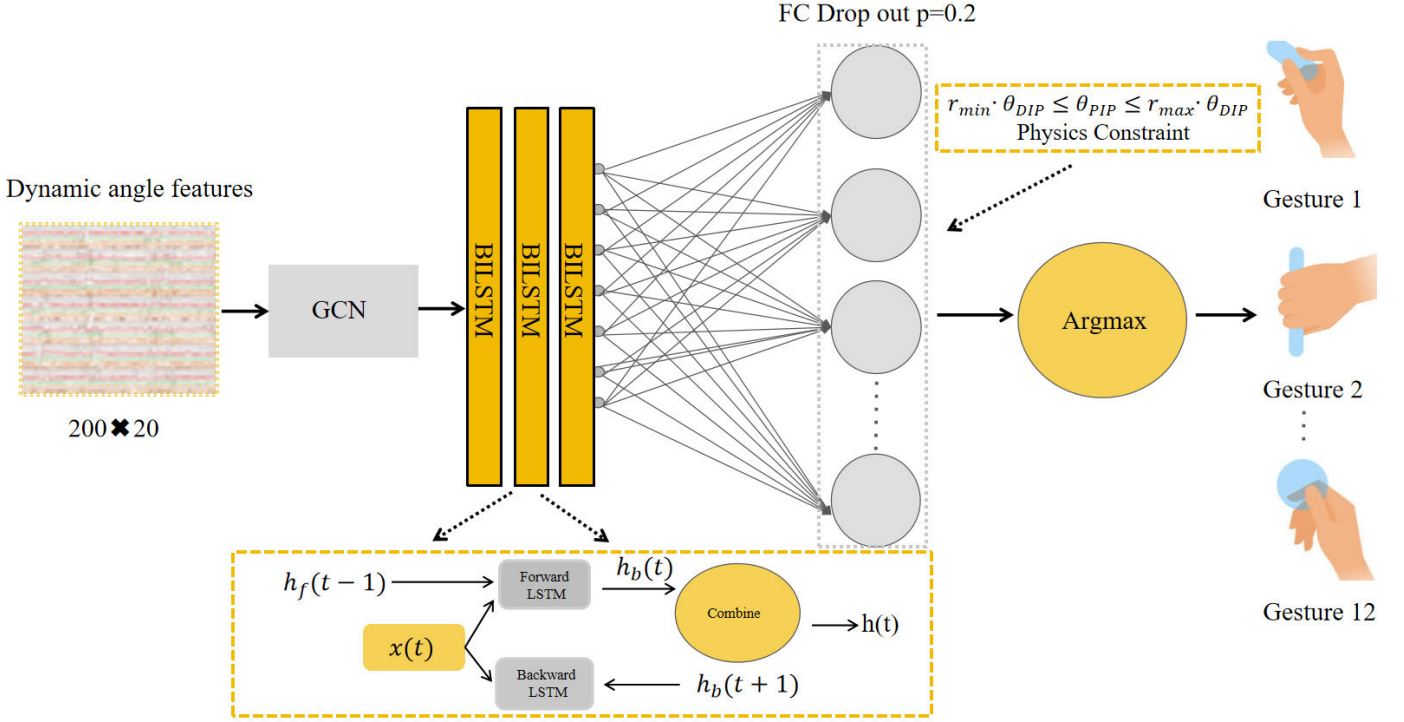


Fig. 5. Schematic of the proposed AGCN-PC model is presented. This model captures finger collaboration relationships through graph convolution, captures long-term dependencies in sequences through a bidirectional LSTM structure, introduces physiological constraints during training, and enhances features.

exhibits complex nonlinear dynamics and varies across individuals.

We leverage this prior knowledge to guide our model. Instead of enforcing a rigid linear equation, we formulate this constraint as a “physiologically plausible range” bounded by anatomical limits. For a given finger j and at a specific time step t , we denote the angles of the PIP and DIP joints as $\theta_{P,j}(t)$ and $\theta_{D,j}(t)$, respectively. We hypothesize their relationship is bounded by a physiologically plausible ratio range, $[r_{j,\min}, r_{j,\max}]$, leading to the following inequalities:

$$\theta_{P,j}(t) \geq r_{j,\min} \cdot \theta_{D,j}(t) \quad (2)$$

$$\theta_{P,j}(t) \leq r_{j,\max} \cdot \theta_{D,j}(t). \quad (3)$$

To enforce these constraints, we define a penalty function for any violation. Let $v_j^{\text{lower}}(t)$ and $v_j^{\text{upper}}(t)$ be the violations of the lower and upper bounds for finger j at time t , respectively,

$$v_j^{\text{lower}}(t) = \max(0, r_{j,\min} \cdot \theta_{D,j}(t) - \theta_{P,j}(t)) \quad (4)$$

$$v_j^{\text{upper}}(t) = \max(0, \theta_{P,j}(t) - r_{j,\max} \cdot \theta_{D,j}(t)). \quad (5)$$

The total physics-based penalty loss, which we term the coupling loss $\mathcal{L}_{\text{couple}}$, is then formulated as the sum of these violations over all relevant fingers $j \in J_{\text{fingers}}$ and all time steps $t \in T$ in the sequence

$$\mathcal{L}_{\text{couple}} = \sum_{t=1}^T \sum_{j \in J_{\text{fingers}}} (v_j^{\text{lower}}(t) + v_j^{\text{upper}}(t)). \quad (6)$$

The final training objective of AGCN-PC is a weighted sum of the standard classification loss, $\mathcal{L}_{\text{class}}$, and our proposed physical constraint loss. The total loss function $\mathcal{L}_{\text{total}}$ is given by the following equation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{class}} + \lambda_{\text{couple}} \mathcal{L}_{\text{couple}} \quad (7)$$

where λ_{couple} is a nonnegative hyperparameter that balances the tradeoff between classification accuracy and adherence to the biomechanical constraints. By minimizing this composite loss, the model is strongly encouraged to learn representations that are not only discriminative for the gesture recognition task but also conform to the physical plausibility of human hand motion, thereby significantly improving its robustness and generalization capabilities.

III. EXPERIMENT

A. Dataset and Acquisition Protocol

The data was collected from 8 volunteers, including 4 males and 4 females. The schematic of the data acquisition process is illustrated in Fig. 6. Among them, 4 participants used their right hand and four participants used their left hand, providing a diverse set of hand characteristics for robust validation. The demographic statistics are as follows: the average age is 27 ± 3.1 years, the average height is 171.6 ± 7.4 cm, and the average weight is 65.5 ± 9.6 kg. The anthropometric data of all eight participants are recorded in Table II to ensure the glove’s adaptability and to validate the system’s generalizability across diverse users. This study was approved by the Medical Ethics Committee of Shenzhen Institutes of Advanced Technology (SIAT-YSB-2022-Y02099).

A total of 12 dynamic hand gestures were designed for the dataset, as illustrated in Fig. 7. These gestures were carefully selected to encompass a wide range of daily activities, covering both power grasps and precision grasps to ensure diversity in finger coordination and joint flexion levels [33]. Specifically, this gesture set ranges from gross power grasps to intricate fine manipulations, effectively covering the core kinematic

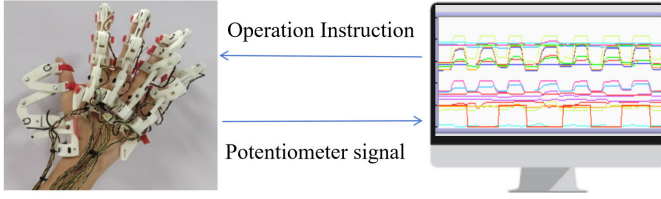


Fig. 6. Schematic of data acquisition process.

TABLE II
DEMOGRAPHIC INFORMATION OF THE EIGHT PARTICIPANTS

Subject ID	Gender	Age (years)	Height (cm)	Weight (kg)	Hand
01	Male	28	175	72	Right
02	Male	24	182	78	Right
03	Female	26	165	58	Right
04	Female	30	160	55	Right
05	Male	32	178	80	Left
06	Male	25	170	68	Left
07	Female	22	170	53	Left
08	Female	29	163	60	Left



Fig. 7. Examples of 12 gestures collected using this exoskeleton glove, demonstrating its capability to capture common hand movements.

primitives required for diverse HCI tasks and functional tool usage scenarios. The schematic of the data acquisition process is illustrated. During data acquisition, participants were seated comfortably. A graphical user interface prompted them to perform a specific gesture. Each gesture trial involved a complete dynamic process: starting from a natural relaxed state, transitioning to the target gesture pose, holding it for a few seconds, and then returning to the relaxed state. The total duration for each trial was approximately 25 s. Each of the 12 gestures was repeated ten times by each participant, resulting in a total of 960 gesture samples. The 15-channel raw angle data were acquired using a National Instruments CompactDAQ system at a sampling rate of 10 Hz.

B. Data Preprocessing

The raw data from the sensors contains minor noise. To address this, we applied a fourth-order low-pass Butterworth filter with a cutoff frequency of 5 Hz. This filtering process effectively removes high-frequency noise while preserving the essential dynamic characteristics of the hand movements, which typically occur below this frequency.

Following filtering, each 25-s trial was segmented to isolate the dynamic gesture instance. These segmented instances were then uniformly resampled to a fixed length of $T = 200$ time steps to ensure consistent input dimensionality for our models. As detailed in our model description, in addition to the 15 base joint angles, we engineered five aggregated features by summing the joint angles for each finger. This resulted in a final feature vector of 20 dimensions for each time step. For all experiments, 30% of the training data in each fold was randomly set aside as a validation set to monitor for overfitting and for hyperparameter tuning.

C. Evaluation Index

1) *Hardware Tracking Performance*: To quantify the ability of our exoskeleton glove to accurately track desired finger movements, we evaluated the fidelity between the measured angle sequences and the predefined target trajectories. Three metrics were used as follows.

- 1) *Root Mean Square Error (RMSE)*: This metric measures the average magnitude of the error between the measured angle θ_m and the desired angle θ_d over T time steps. It is sensitive to large errors

$$RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T (\theta_m(t) - \theta_d(t))^2}. \quad (8)$$

- 2) *Mean Absolute Error (MAE)*: This metric provides the average absolute difference, giving an intuitive measure of the overall tracking error

$$MAE = \frac{1}{T} \sum_{t=1}^T |\theta_m(t) - \theta_d(t)|. \quad (9)$$

- 3) *Pearson Correlation Coefficient (r)*: This metric assesses the linear relationship and temporal similarity between the two sequences, ranging from -1 to 1 . A value close to 1 indicates a near-perfect match in the dynamic pattern of the movement.

2) *Model Recognition Performance*: For the 12-class gesture recognition task, we used four standard classification metrics derived from the confusion matrix. Given a multiclass scenario, these metrics are typically computed on a one-versus-rest basis for each class i and then averaged across all classes. Let TP_i , FP_i , TN_i , and FN_i be the number of true positives, false positives, true negatives, and false negatives for class i .

- 1) *Accuracy*: The proportion of all correctly classified samples

$$\text{Accuracy} = \frac{\sum_{i=1}^C TP_i}{\text{Total Samples}} \quad (10)$$

where C is the number of classes ($C = 12$).

TABLE III

QUANTITATIVE EVALUATION OF DYNAMIC TRACKING PERFORMANCE FOR THE EXOSKELETON GLOVE. THE “–” SYMBOL INDICATES THAT THE FINGER WAS NOT ACTIVELY INVOLVED IN THE GESTURE SHOWN IN THE CORRESPONDING FIGURE

Gesture/Sequence	Metric	Thumb (°)	Index (°)	Middle (°)	Ring (°)	Little (°)	Avg./Overall (°)
Single Gestures							
Thumb Opposition (Fig. 9.b)	RMSE	6.5	–	–	–	–	6.5
	MAE	4.8	–	–	–	–	4.8
	r	0.993	–	–	–	–	0.993
Tip Pinch (Fig. 9.c)	RMSE	7.2	7.8	–	–	–	7.5
	MAE	5.5	6.0	–	–	–	5.8
	r	0.990	0.994	–	–	–	0.992
Lateral Pinch (Fig. 9.d)	RMSE	8.0	8.8	7.5	7.2	7.0	7.7
	MAE	6.1	6.7	5.8	5.5	5.3	5.9
	r	0.987	0.992	0.980	0.978	0.975	0.982
Fist (Fig. 9.e)	RMSE	7.9	7.5	8.1	8.3	8.5	8.1
	MAE	6.0	5.8	6.2	6.4	6.6	6.2
	r	0.991	0.993	0.990	0.988	0.987	0.990
Sequential Gesture Transition							
Tip Pinch → Fist → Tip Pinch (Fig. 10)	RMSE	9.2	8.9	7.5	7.2	4.5	7.5
	MAE	7.0	6.8	5.7	5.4	3.2	5.6
	r	0.983	0.986	0.978	0.976	0.960	0.977

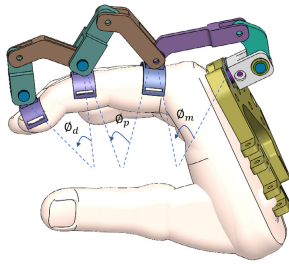


Fig. 8. Definition of finger flexion degree, the precision tracking index for the exoskeleton.

- 2) *Precision*: The average ability of the classifier not to label a negative sample as positive

$$\text{Precision} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FP_i}. \quad (11)$$

- 3) *Recall*: Also known as sensitivity, it is the average ability of the classifier to find all the positive samples

$$\text{Recall} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FN_i}. \quad (12)$$

- 4) *F1-Score*: The harmonic mean of precision and recall, providing a single score that balances both metrics

$$\text{F1-Score} = \frac{1}{C} \sum_{i=1}^C 2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}. \quad (13)$$

IV. RESULTS AND DISCUSSION

A. Hardware Performance Validation: Dynamic Tracking Fidelity

To validate that our low-cost exoskeleton glove can serve as a high-fidelity measurement instrument, we first needed to define a comprehensive metric for finger movement. For the

purpose of evaluating dynamic tracking, we define the total flexion degree of a single finger, denoted by Φ , as the sum of its three constituent joint angles: the Metacarpophalangeal, Proximal Interphalangeal, and Distal Interphalangeal joints. This composite metric, illustrated in Fig. 8, provides a holistic measure of the overall bending state of a finger

$$\Phi = \phi_d + \phi_p + \phi_m. \quad (14)$$

All subsequent tracking performance evaluations are presented in Figs. 9, 10, and Table III utilize this total flexion degree as the primary tracking index for each finger.

1) *Tracking of Single Gestures*: First, we evaluated the glove's ability to track four primary gestures: “thumb opposition,” “tip pinch,” “lateral pinch,” and “fist,” as defined in Fig. 9(a). The dynamic tracking results are visualized in Fig. 9(b)–(e), where the solid lines represent the measured angles and the dashed lines represent the target trajectories. Visually, the measured curves demonstrate excellent alignment with the target curves. The quantitative results for these single gestures are detailed in the upper section of Table III. For all evaluated gestures, the system achieved exceptionally high temporal correlation, with Pearson's r values consistently exceeding 0.98. For instance, during the “fist” gesture, where all five fingers are actuated, the average correlation coefficient is 0.990. The average RMSE across all fingers for this gesture is 8.1° , a low value relative to the large range of motion involved. For gestures involving fewer fingers, such as “tip pinch,” the tracking fidelity remains excellent ($r = 0.992$), showcasing the glove's precision in capturing both power and precision grasps. The table's use of “–” indicates fingers that were not part of the target motion, confirming no significant crosstalk or unintended measurement.

2) *Tracking of Sequential Gesture Transitions*: To assess the system's performance in a more challenging and realistic scenario, we designed a sequential transition task: “tip pinch” → “fist” → “tip pinch.” This task requires the glove to

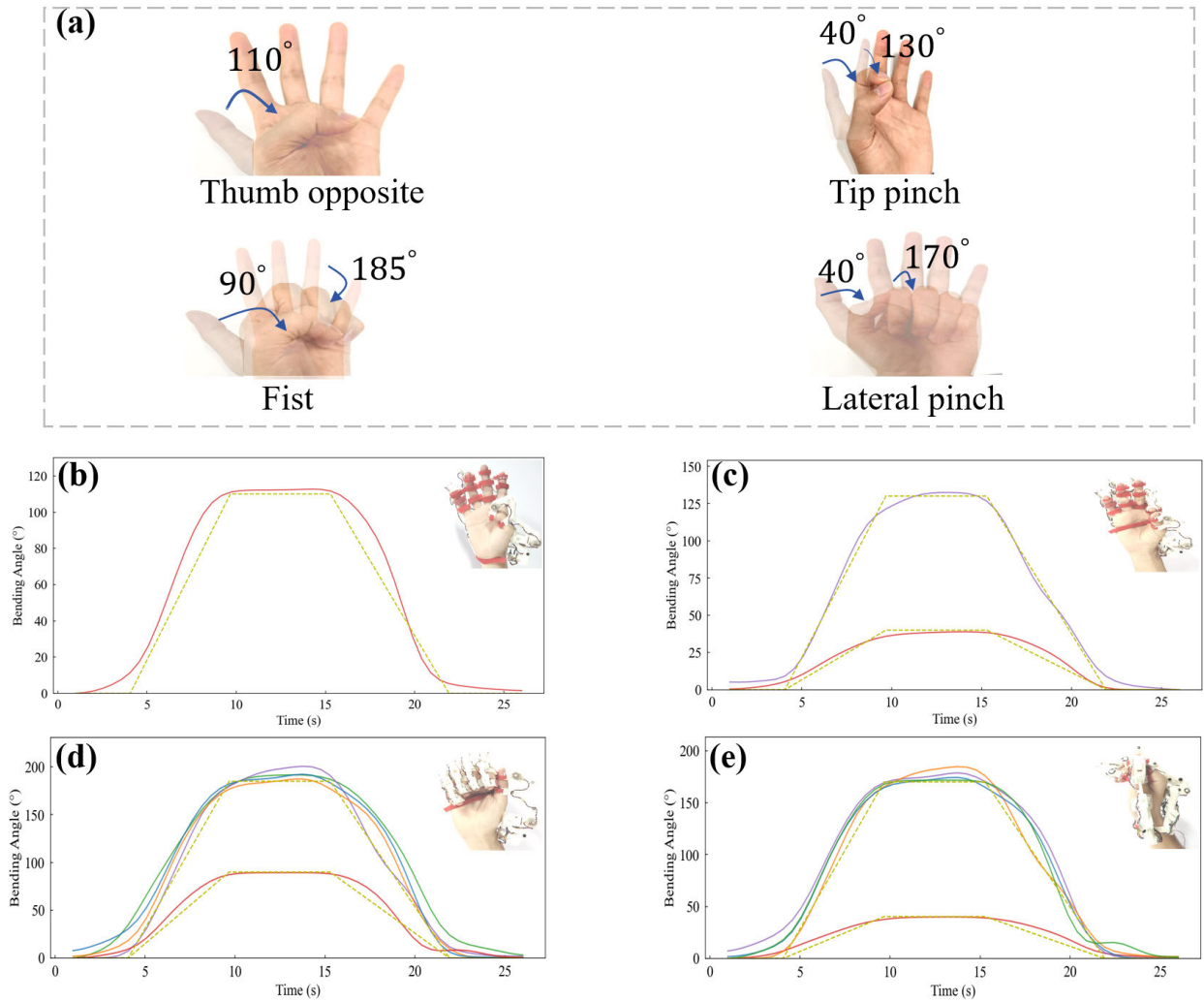


Fig. 9. Visualization of four primary gestures and their corresponding dynamic tracking performance. (a) Target static poses. (b)–(e) Tracking curves for each single gesture, where the inset images show the hardware posture.

track rapid changes in coordination patterns across all five fingers. The continuous tracking performance for this task is illustrated in Fig. 10. The plot clearly shows that the glove accurately captures the complex dynamics, including the distinct motion profiles of each finger as they transition between a precision grip and a power grip. The quantitative analysis for this sequence, presented in the lower section of Table III, further corroborates this high performance. The overall Pearson correlation coefficient remains very high at 0.977, and the average RMSE is only 7.5° . This demonstrates the system's ability to maintain high tracking fidelity even during complex, dynamic, and multistage movements.

The high correlation coefficient not only demonstrates tracking accuracy but also validates the mechanical stability of the rigid linkage design, confirming that the exoskeleton follows the finger motion without significant hysteresis or mechanical play.

B. Model Classification Effect Test

1) Performance Comparison Against Baseline Methods:

In this experiment, we benchmarked our AGCN-PC model

TABLE IV
DETAILED RECOGNITION PERFORMANCE ON GESTURE DATASET

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
LSTM	77.8	78.5	77.8	78.0
CNN	81.5	82.1	81.5	81.7
BiGRU	83.5	84.1	83.0	83.8
BiLSTM	83.9	84.4	83.9	84.0
CNN+LSTM	90.3	91.0	90.3	90.5
CNN-BiLSM	94.2	94.6	94.2	94.4
AGCN-PC	98.5	98.8	98.5	98.6

against several strong baseline methods. We adopt the same data preprocessing method and the same dataset division approach to compare the model performance. The recognition accuracy performance of each model is plotted in Fig. 11. The detailed performance metrics, including accuracy, precision, recall, and $F1$ -Score, are presented in Table IV.

The results demonstrate that our proposed AGCN-PC model achieves a perfect score of 98.5% across all evaluation metrics. This performance represents a significant margin over all competing methods. For instance, the next-best performing

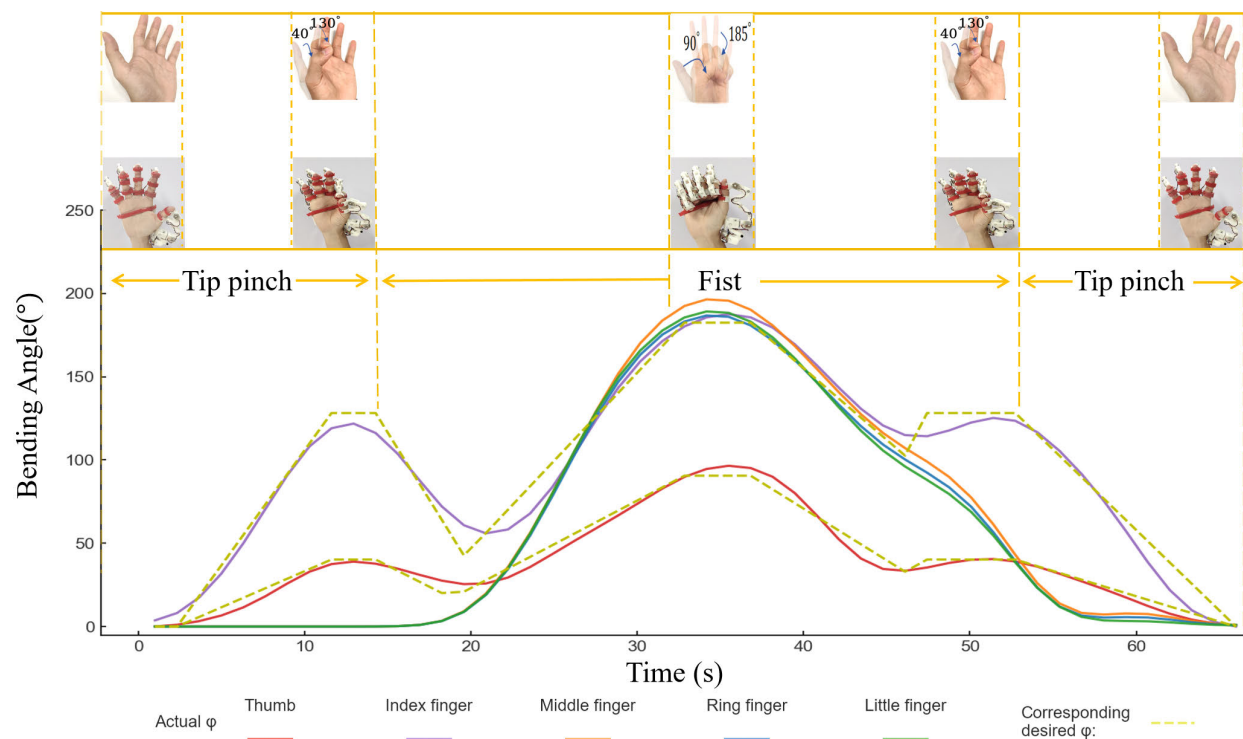


Fig. 10. Dynamic tracking performance during a sequential gesture transition task: "tip pinch"—"fist"—"tip pinch."

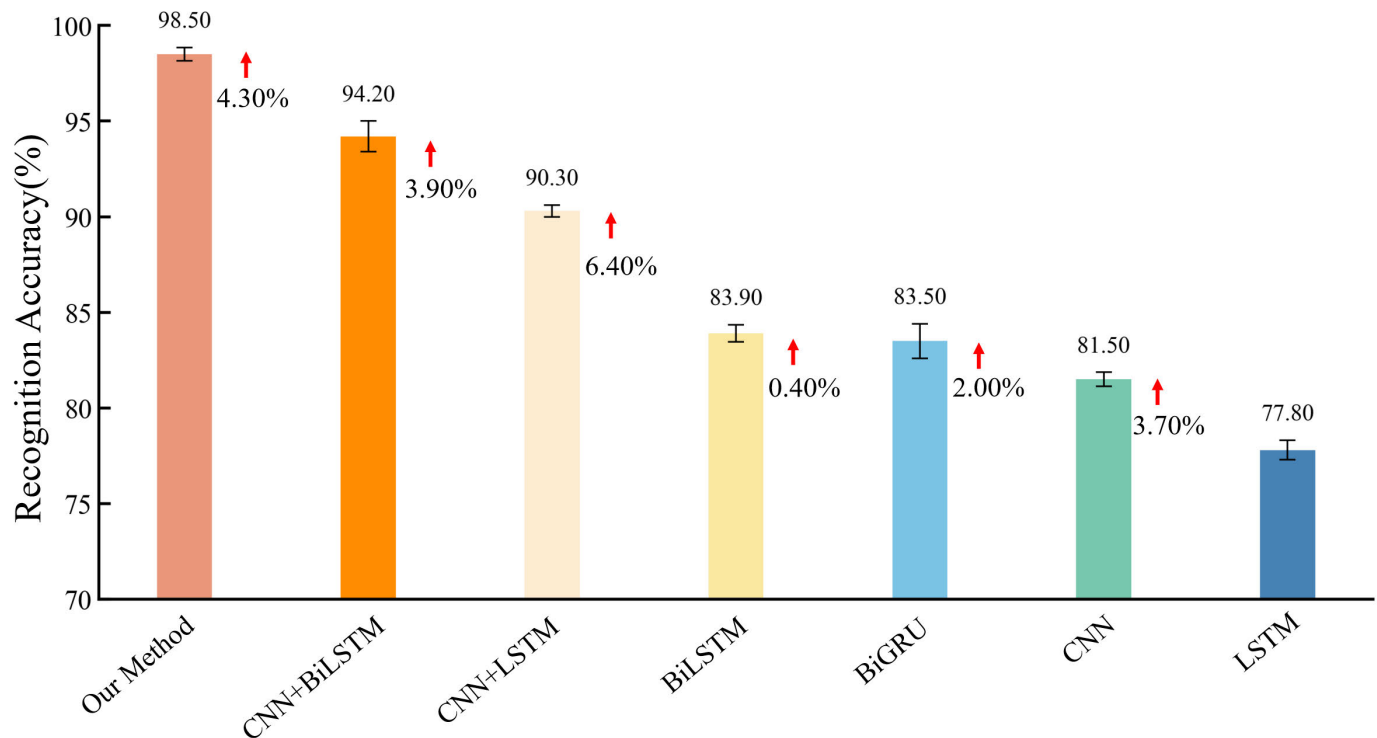
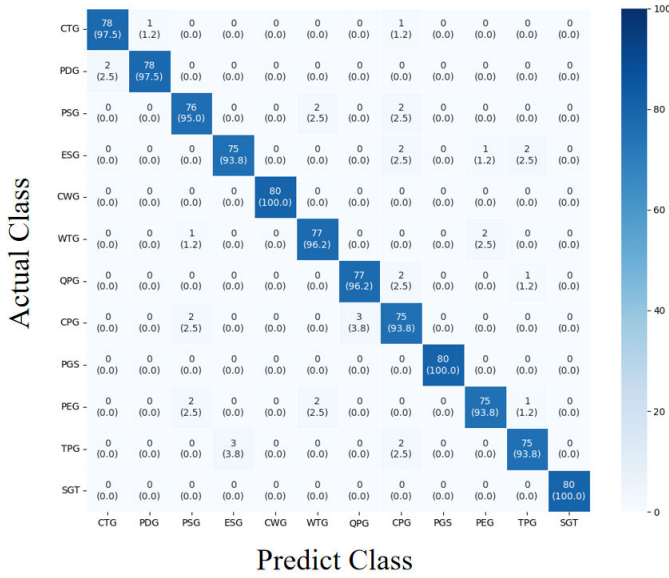


Fig. 11. Performance comparison of AGCN-PC against other benchmark deep learning architectures dataset. The bars represent the mean accuracy across all subjects, with error bars indicating the standard deviation.

model, CNN-BiLSTM, achieved an accuracy of only 94.2%, while another powerful hybrid model, CNN + LSTM, reached just 90.3%. These results decisively establish the superiority of our proposed architecture.

2) *Intersubject Validation*: To rigorously evaluate the model's generalization capability to new, unseen users, we employed a strict leave-one-out cross-validation. Our dataset, comprising 960 samples from eight subjects, was partitioned



Predict Class

Fig. 12. Overall confusion matrix for the 12-gesture classification task, derived from the leave-one-out cross-validation across eight subjects. Each cell displays the instance count and the corresponding row-wise percentage.

TABLE V

LOOCV RESULTS FOR EIGHT SUBJECTS. THE TABLE SHOWS PER-FOLD PERFORMANCE AND OVERALL STATISTICS

Fold / Statistic	Test Subject	Accuracy (%)	F1-Score (%)
1	Subject 1	95.00	95.0
2	Subject 2	96.67	96.7
3	Subject 3	95.00	95.1
4	Subject 4	96.67	96.7
5	Subject 5	95.25	95.2
6	Subject 6	94.75	94.8
7	Subject 7	96.25	96.3
8	Subject 8	95.50	95.6
Average	—	95.64	95.68
Std. Dev.	—	0.70	0.71

into eight folds, where each fold contained all 120 samples from a single subject. In each iteration, one subject's data was used for testing, while the data from the remaining three subjects were used for training. This protocol directly simulates the real-world scenario of a new user interacting with a pretrained system. The detailed results of this demanding validation are presented in Table V. Our AGCN-PC model demonstrated both high accuracy and remarkable consistency. The model achieved a mean accuracy of 95.64% and a mean $F1$ -Score of 95.68%. More importantly, the low standard deviation across the folds (0.70 for accuracy and 0.71 for $F1$ -score) highlights the model's robustness to intersubject variations. The performance remained stable regardless of which subject was held out for testing, with individual accuracies ranging narrowly from 95.00% to 96.67%. This high and stable performance strongly validates the ability of our physics-informed model to learn user-agnostic gesture representations, a critical requirement for practical, deployable systems. To provide a class-by-class breakdown of the model's cross-user generalization performance, we present the overall confusion matrix and the ROC curves aggregated from all LOOCV folds. As

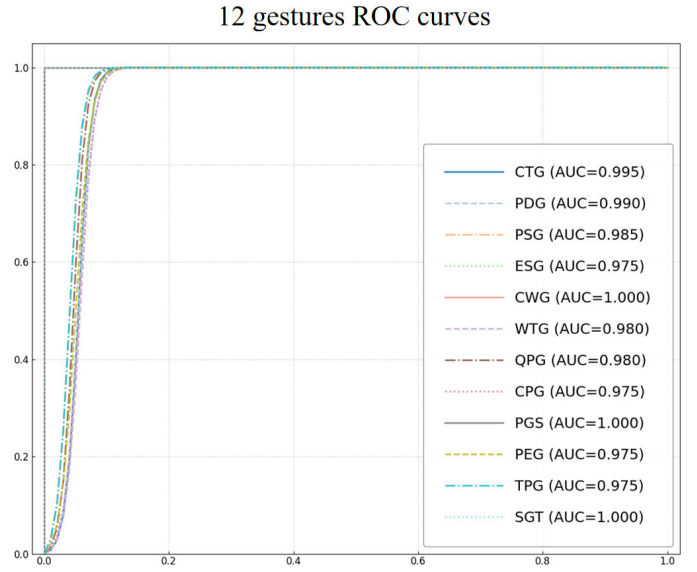


Fig. 13. One-versus-rest ROC curves for 12-class gesture recognition.

shown in Fig. 12. The matrix uses three-letter abbreviations for each gesture, as defined in Fig. 7. The strong diagonal pattern, with most cells exceeding 95% recall, confirms the model's excellent discriminative power across the majority of gestures. Notably, several challenging fine-motor gestures, such as WTG and TPG, were recognized with high accuracy, demonstrating the system's ability to capture subtle finger coordinations even across different users. Furthermore, gestures like CWG and PGS also achieved near-perfect recognition, showcasing robustness in identifying common power grasps.

The matrix also reveals that the few misclassifications primarily occur between gestures that are posturally similar and can be challenging even for human observation to distinguish. The most significant confusion is observed for TPG, which has the lowest recall of 93.8%. It is occasionally misclassified as ESG (and, more frequently, as CPG). This is plausible, as these gestures share similar foundational hand shapes before interacting with an object. Similarly, a minor confusion exists between CPG and PSG. These subtle errors highlight the inherent difficulty of the task. Nevertheless, the overwhelmingly high accuracy across the board robustly validates that our AGCN-PC model, informed by physical priors, can effectively learn generalizable representations for a complex and diverse set of hand gestures. As shown in Fig. 13, the ROC curves for the 12 gestures on the test set demonstrate excellent performance, with high AUC values for each gesture. These values indicate that the model effectively differentiates between the positive and negative classes for each gesture.

3) *Data Efficiency and Learning Curve Analysis*: To further investigate the robustness and remarkable data efficiency of the AGCN-PC model, we conducted a comprehensive learning curve analysis. This analysis was performed under a robust pooled-subject, fivefold cross-validation setup, evaluating model performance under varying data volumes. Specifically, all models were trained on randomly selected subsets corresponding to 33%, 67%, and 100% of the total available training samples. This process was repeated five

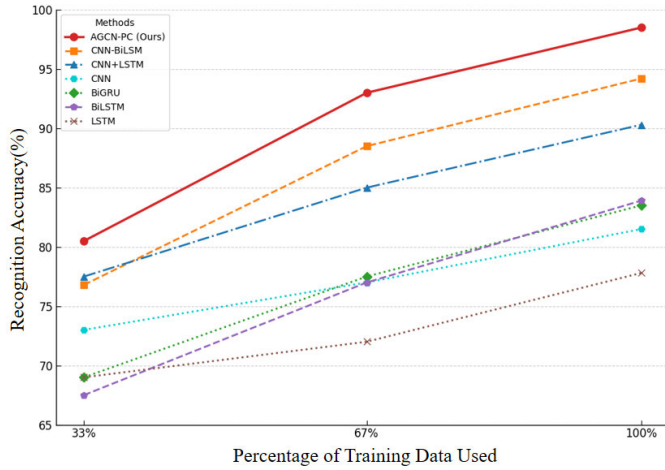


Fig. 14. Data efficiency comparison of AGCN-PC against baseline models.

TABLE VI
ABLATION STUDY OF THE PROPOSED AGCN-PC COMPONENTS

Model ID	Graph Structure	Coupling Constraints	Accuracy (%)
M1	Generic [34]		94.1
M2	Ours		96.0
M3	Ours	✓	98.5

times for statistical reliability, and the resulting mean accuracy curves are presented in Fig. 14.

As illustrated in the figure, our physics-informed model consistently and significantly outperforms all benchmark methods across all data regimes. While all models show expected performance improvements with increasing data, the decisive advantage of our physics-informed approach is most pronounced in the data-scarce regime.

Most notably, with only 33% of the training data, our AGCN-PC model achieves an accuracy of 80.50%. In contrast, the second-best performing model at this low-data level, CNN-LSTM, only reaches 77.50%, yielding a substantial performance lead of 3.00% points for our model.

As the training data increases, our AGCN-PC model consistently maintains a significant performance gap. At 100% data utilization, it achieves an accuracy of 98.50%, demonstrating its superior generalization capability on the full, diverse dataset. This comprehensive analysis confirms that the synergy of affordable hardware and physics-informed AI enables remarkable data efficiency, making it an ideal solution for resource-constrained practical applications where collecting large-scale labeled datasets is costly or impractical.

4) *Ablation Study of Model Components*: To dissect the sources of this superior performance, we conducted an ablation study designed to quantify the contribution of our two main innovations: the high-fidelity graph structure and the interjoint coupling constraints. The results are summarized in Table VI. The importance of the graph structure is evident when comparing the first two models in the study. A baseline model using a generic, anatomically based on hand graph (M1) achieved an accuracy of 94.1%. By replacing this generic graph with our proposed high-fidelity graph derived

from the exoskeleton's physics (M2), the accuracy surged to 96.0%. This 1.9% point increase confirms that encoding precise physical kinematics into the graph structure is a fundamental and highly effective strategy. Building upon this strong foundation, we then evaluated the effect of our physical constraints. The addition of the interjoint coupling constraints to the M2 model resulted in our full AGCN-PC model (M3). This single addition further boosted the performance from an already high 96.0% to a perfect accuracy of 98.5%. This final 2.5% improvement underscores the critical role of the physical constraints, which act as a powerful regularizer, guiding the model to eliminate ambiguities between similar gestures and learn more physically plausible representations.

V. CONCLUSION

In this article, we have presented a novel, integrated system that resolves the long-standing tradeoff between cost, accuracy, and robustness in hand gesture recognition. Our work validates the principle that a hardware–software co-design philosophy can unlock high-performance gesture recognition using exceptionally low-cost components. At the core of our system is a 3-D-printed exoskeleton glove, developed for under U.S. \$20, which functions as a high-fidelity measurement instrument for capturing 15-DoF hand kinematics with high repeatability. This hardware foundation is leveraged by our proposed physics-constrained graph neural network (GNN), AGCN-PC, which uniquely embeds the glove's mechanical structure as a graph prior and incorporates biomechanical coupling constraints. Validated through comprehensive experiments, this synergistic approach achieves a remarkable 95.64% accuracy in challenging cross-user scenarios and 98.50% in user-specific tests, significantly outperforming all baselines. Ultimately, the model's exceptional data efficiency underscores the profound impact of embedding physical knowledge into AI architectures, paving the way for accessible, high-precision HCI.

Looking forward, our work opens several promising avenues for future research. First, to facilitate the transition from a research prototype to a consumer-grade interface, future hardware iterations will prioritize ergonomic optimization for extended wearability and undergo rigorous long-term fatigue testing to ensure the durability standards required for daily HCI and VR applications. Furthermore, we intend to integrate additional sensing modalities, such as thin-film pressure sensors, to enable grasp force estimation. Second, the AGCN-PC model could be extended to incorporate more complex, nonlinear biomechanical constraints or adaptive, user-specific constraints that are learned online. Finally, deploying and evaluating the system in real-world clinical or interactive scenarios will be a crucial next step to validate its practical utility and impact. We believe this synergy of intelligent mechanical design and physics-informed AI represents a powerful paradigm for the future of HCI.

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